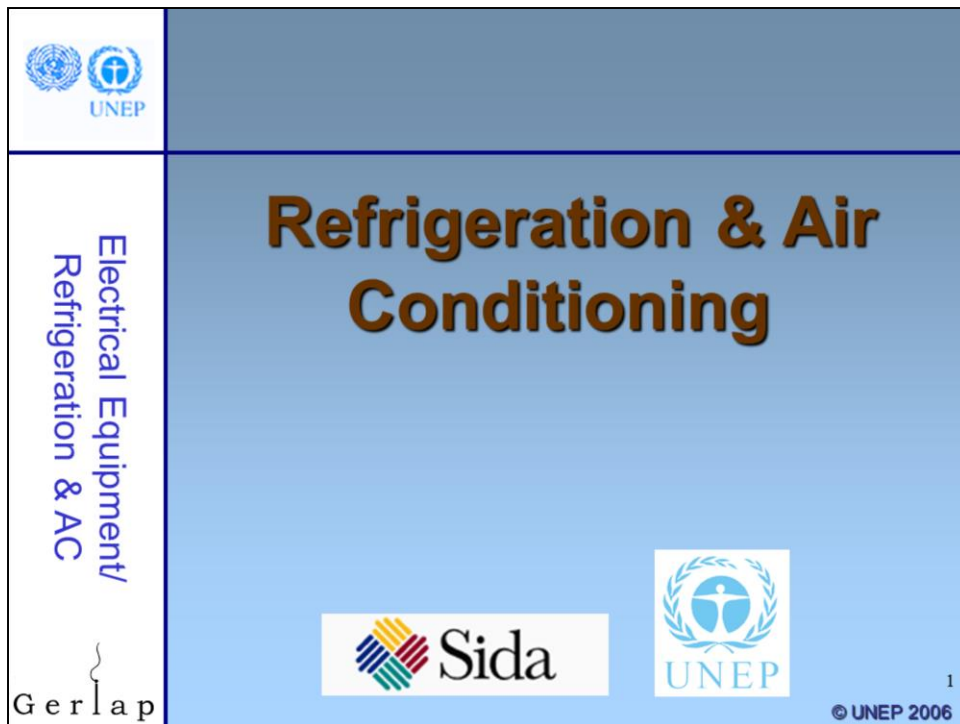




TO THE TRAINER

This PowerPoint presentation can be used to train people about the basics of refrigeration and air conditioning. The information on the slides is the minimum information that should be explained. The trainer notes for each slide provide more detailed information, but it is up to the trainer to decide if and how much of this information is presented also.

Additional materials that can be used for the training session are available on www.energyefficiencyasia.org under “Energy Equipment” and include:

- **Textbook chapter** on this energy equipment that forms the basis of this PowerPoint presentation but has more detailed information
- **Quiz** – ten multiple choice questions that trainees can answer after the training session
- **Option checklist** – a list of the most important options to improve energy efficiency of this equipment

 UNEP	Introduction
Electrical Equipment/ Refrigeration & AC Gerlap	Refrigeration systems for industrial processes • Small capacity modular units of direct expansion type (50 Tons of Refrigeration) • Centralized chilled water plants with chilled water as a secondary coolant (50 – 250 TR) • Brine plants with brines as lower temperature, secondary coolant (>250 TR) 2 © UNEP 2006

The following refrigeration systems exist for industrial processes (e.g. chilling plants) and domestic purposes (modular units, i.e. refrigerators):

- Small capacity modular units of the direct expansion type similar to domestic refrigerators.

- Centralized chilled water plants with chilled water as a secondary coolant for a temperature range over typically 5 oC. They can also be used for ice bank formation.

- Brine plants, which use brines as a lower temperature, secondary coolant for typically sub-zero temperature applications, which come as modular unit capacities as well as large centralized plant capacities.

The plant capacities up to 50 TR (tons of refrigeration) are usually considered as small capacity, 50 – 250 TR as medium capacity and over 250 TR as large capacity units.

 UNEP	Types of Refrigeration
Electrical Equipment/ Refrigeration & AC 	Refrigeration systems • Vapour Compression Refrigeration (VCR): uses mechanical energy • Vapour Absorption Refrigeration (VAR): uses thermal energy 3 © UNEP 2006

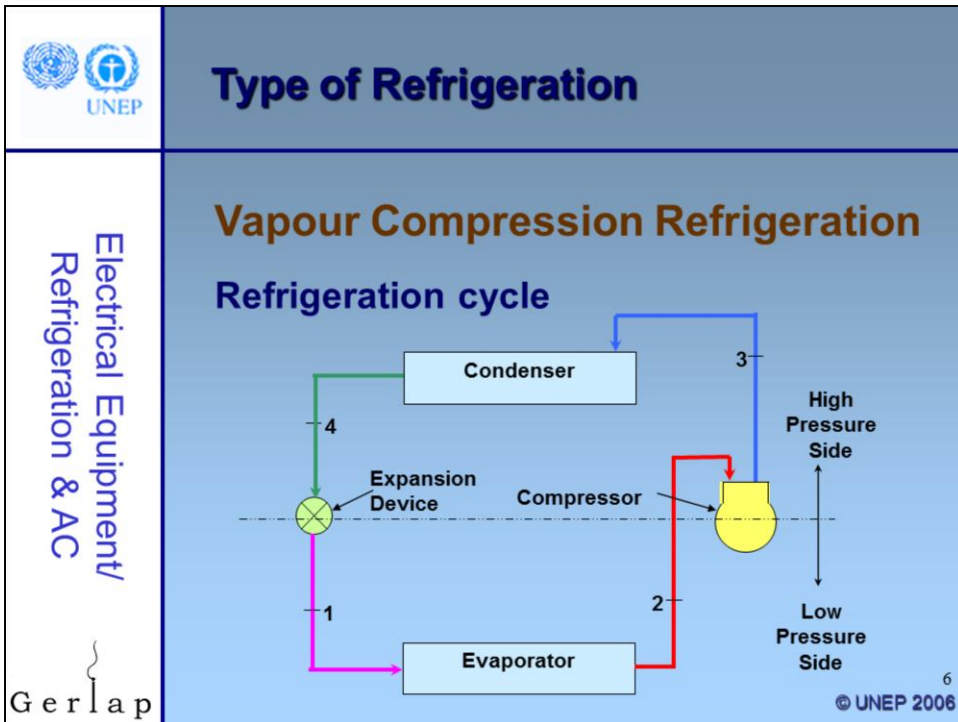
- The two principle types of refrigeration plants found in industrial use are: Vapour compression refrigeration (VCR) and vapour absorption refrigeration (VAR). VCR uses mechanical energy as the driving force for refrigeration, while VAR uses thermal energy as the driving force for refrigeration.

 UNEP	Type of Refrigeration
Electrical Equipment/ Refrigeration & AC Gerlap	Vapour Compression Refrigeration • Highly compressed fluids tend to get colder when allowed to expand • If pressure high enough • Compressed air hotter than source of cooling • Expanded gas cooler than desired cold temperature 4 © UNEP 2006

- Compression refrigeration cycles take advantage of the fact that highly compressed fluids at a certain temperature tend to get colder when they are allowed to expand.
- If the pressure change is high enough, then the compressed gas will be hotter than our source of cooling (outside air, for instance) and the expanded gas will be cooler than our desired cold temperature. In this case, fluid is used to cool a low temperature environment and reject the heat to a high temperature environment.

 UNEP	Type of Refrigeration
Electrical Equipment/ Refrigeration & AC Gerlap	Vapour Compression Refrigeration Two advantages • Lot of heat can be removed (lot of thermal energy to change liquid to vapour) • Heat transfer rate remains high (temperature of working fluid much lower than what is being cooled) 5 © UNEP 2006

- Vapour compression refrigeration cycles have two advantages.
 - First, a large amount of thermal energy is required to change a liquid to a vapor, and therefore a lot of heat can be removed from the air-conditioned space.
 - Second, the isothermal nature of the vaporization allows extraction of heat without raising the temperature of the working fluid to the temperature of whatever is being cooled. This means that the heat transfer rate remains high, because the closer the working fluid temperature approaches that of the surroundings, the lower the rate of heat transfer.



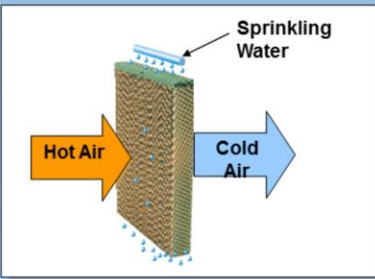
The refrigeration cycle is shown in the Figure and can be broken down into the following stages (*note to the trainer: the next slides will explain what is happening between 1 to 4*)

 UNEP	Type of Refrigeration
Electrical Equipment/ Refrigeration & AC 	Vapour Compression Refrigeration Type of refrigerant • Refrigerant determined by the required cooling temperature • Chlorinated fluorocarbons (CFCs) or freons: R-11, R-12, R-21, R-22 and R-502 7 © UNEP 2006

- A variety of refrigerants are used in vapor compression systems. The choice of fluid is determined largely by the cooling temperature required.
- Commonly used refrigerants are in the family of chlorinated fluorocarbons, CFCs, also known as freons: R-11, R-12, R-21, R-22 and R-502.

 UNEP	Type of Refrigeration
Electrical Equipment/ Refrigeration & AC  Gerlap	Vapour Compression Refrigeration Choice of compressor, design of condenser, evaporator determined by • Refrigerant • Required cooling • Load • Ease of maintenance • Physical space requirements • Availability of utilities (water, power) 8 © UNEP 2006

- The choice of refrigerant and the required cooling temperature and load determine the choice of compressor, as well as the design of the condenser, evaporator, and other auxiliaries.
- Additional factors such as ease of maintenance, physical space requirements and availability of utilities for auxiliaries (water, power, etc.) also influence component selection.

 UNEP	Type of Refrigeration
Electrical Equipment/ Refrigeration & AC Gerlap	Evaporative Cooling • Air in contact with water to cool it close to 'wet bulb temperature' • Advantage: efficient cooling at low cost • Disadvantage: air is rich in moisture  (Adapted from Munters, 2001) 9 © UNEP 2006

- There are occasions where air conditioning, which stipulates control of humidity up to 50 % for human comfort or for process, can be replaced by a much cheaper and less energy intensive evaporative cooling. Evaporative cooling is an extremely efficient means of cooling at very low cost.
- The concept is very simple and is the same as that used in a cooling tower. Air is brought in close contact with water to cool it to a temperature close to the wet bulb temperature. The cool air can be used for comfort or process cooling.
- The disadvantage is that the air is rich in moisture.
- The possibility of evaporative cooling is especially attractive for comfort cooling in dry regions. This principle is practiced in textile industries for certain processes.

 UNEP	Assessment of Refrigeration and AC
Electrical Equipment/ Refrigeration & AC 	Assessment of Refrigeration • Cooling effect: Tons of Refrigeration 1 TR = 3024 kCal/hr heat rejected • TR is assessed as: $TR = Q \times C_p \times (T_i - T_o) / 3024$ Q = mass flow rate of coolant in kg/hr C_p = is coolant specific heat in kCal /kg deg C T_i = inlet, temperature of coolant to evaporator (chiller) in 0C T_o = outlet temperature of coolant from evaporator (chiller) in 0C 10 © UNEP 2006

- The cooling effect produced is quantified as tons of refrigeration, or shortly TR. 1 TR of refrigeration = 3024 kCal/hr heat rejected.
- The refrigeration TR is assessed as $TR = Q \times C_p \times (T_i - T_o) / 3024$, where Q is mass flow rate of coolant; C_p is coolant specific heat in; T_i is inlet, temperature of coolant to evaporator; and T_o is outlet temperature of coolant from evaporator. The above TR is also called as chiller tonnage.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	1. Optimize process heat exchange 2. Maintain heat exchanger surfaces 3. Multi-staging systems 4. Matching capacity to system load 5. Capacity control of compressors 6. Multi-level refrigeration for plant needs 7. Chilled water storage 8. System design features 11 © UNEP 2006

We will go through the following eight areas for energy efficiency improvements.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC Gerlap	1. Optimize Process Heat Exchange High compressor safety margins: energy loss Proper sizing heat transfer areas of heat exchangers and evaporators • Heat transfer coefficient on refrigerant side: 1400 – 2800 Watt/m²K • Heat transfer area refrigerant side: >0.5 m²/TR Optimum driving force (difference T_e and T_c): 1°C raise in T_e = 3% power savings 12 © UNEP 2006

- There is a tendency to apply high safety margins to operations, which influence the compressor suction pressure / evaporator set point.
 - For instance, a process-cooling requirement of 15 oC would need chilled water at a lower temperature, but the range can vary from 6 oC to about 10 oC. At chilled water of 10 oC, the refrigerant side temperature has to be lower (about –5oC to +5oC).
 - The refrigerant temperature determines the corresponding suction pressure of the refrigerant, which in turn determines the inlet duty conditions for the refrigerant compressor.

Minimizing energy consumption can be achieved in the following ways:

- Proper sizing of heat transfer areas** of process heat exchangers and evaporators
 - the heat transfer coefficient on the refrigerant side can range from 1400 – 2800 watts /m²K.
 - The refrigerant side heat transfer areas are of the order of 0.5 m²/TR and above in evaporators
- Optimizing the driving force, i.e. the difference between evaporator temp T_e and condensing temp T_c.**
 - A 1oC raise in evaporator temperature can save almost 3 % of the power

consumed.

- The next slide shows the effect of T_e and T_c on the cooling effect TR and the power consumption



UNEP

Energy Efficiency Opportunities

Electrical Equipment/
Refrigeration & AC

Gerlap

1. Optimize Process Heat Exchange

Evaporator Temperature (°C)	Refrigeration Capacity (tons)	Specific Power Consumption (kW/TR)	Increase kW/TR (%)
5.0	67.58	0.81	-
0.0	56.07	0.94	16.0
-5.0	45.98	1.08	33.0
-10.0	37.20	1.25	54.0
-20.0	23.12	1.67	106.0

Condenser temperature 40- C (National Productivity Council)

Condensing Temperature (°C)	Refrigeration Capacity (tons)	Specific Power Consumption (kW/TR)	Increase kW/TR (%)
26.7	31.5	1.17	-
35.0	21.4	1.27	8.5
40.0	20.0	1.41	20.5

*Reciprocating compressor using R-22 refrigerant. Evaporator temperature -10- C

13

© UNEP 2006

- This table shows the effect of evaporation temperature on the compressor power consumption. At a constant condenser temperature T_c of 40°C, lower evaporator temperatures reduce the refrigeration capacity TR and increase the power consumption
- The second table shows the effect of condensing temperature on the compressor power consumption. At a constant evaporator temperature T_e of 10°C, increasing the condensing temperature leads to a reduction in refrigeration capacity and an increase in power consumption.
- The conclusion is therefore that you try to keep the difference between T_e and T_c at an optimum level to ensure the best TR at the lowest power consumption.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC Gerlap	1. Optimize Process Heat Exchange 3. Selection of condensers - Options: - Air cooled condensers - Air-cooled with water spray condensers - Shell & tube condensers with water-cooling - Water-cooled shell & tube condenser - Lower discharge pressure - Higher TR - Lower power consumption 14 © UNEP 2006

3. Selection of condensers

- The choice of condensers in practice is between air-cooled, air-cooled with water spray, and shell and tube condensers with water-cooling.
- Larger shell and tube heat exchangers (0.65 m²/TR and above) that are used as condensers and that are equipped with good cooling tower operations allow operation at low discharge pressure values and improve the TR capacity of the refrigeration plant and thus reduce the power consumption.
- For example (*optional*)
 - If the refrigerant R22 is used in a water-cooled shell and tube condenser then the discharge pressure is 15 kg/cm².
 - If the same refrigerant is used in an air-cooled condenser then the discharge pressure is 20 kg/cm². This shows how much additional compression duty is required, which results in almost 30 % additional energy consumption by the plant.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	2. Maintain Heat Exchanger Surfaces • Poor maintenance = increased power consumption • Maintain condensers and evaporators • Separation of lubricating oil and refrigerant • Timely defrosting of coils • Increased velocity of secondary coolant • Maintain cooling towers • 0.55°C reduction in returning water from cooling tower = 3.0 % reduced power 15 © UNEP 2006

• Once compressors have been purchased, effective maintenance is the key to optimizing power consumption. Poor maintenance forces a compressor to work harder, which results in increased power consumption

• A maintenance checklist for the condenser and evaporator

- Ensuring proper separation of the lubricating oil and the refrigerant
- Timely defrosting of coils
- Increasing the velocity of the secondary coolant (air, water, etc.).

• Equally important is proper selection, sizing, and maintenance of cooling towers. A reduction of 0.55°C in temperature of the water returning from the cooling tower reduces compressor power consumption by 3%.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	3. Multi-Staging Systems • Suited for • Low temp applications with high compression • Wide temperature range • Two types for all compressor types • Compound • Cascade 16 © UNEP 2006

- Efficient compressor operation requires that the compression ratio be kept low, to reduce discharge pressure and temperature. For low temperature applications involving high compression ratios, and for wide temperature range requirements, it is preferable (due to equipment design limitations) and often economical to employ multi-stage reciprocating machines or centrifugal / screw compressors.
- There are two types of multi-staging systems, which are applicable to all types of compressors: compound and cascade.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	3. Multi-Stage Systems a. Compound • Two low compression ratios = 1 high • First stage compressor meets cooling load • Second stage compressor meets load evaporator and flash gas • Single refrigerant b. Cascade • Preferred for -46 oC to -101oC • Two systems with different refrigerants 17 © UNEP 2006

•Efficient compressor operation requires that the compression ratio be kept low, to reduce discharge pressure and temperature. For low temperature applications involving high compression ratios, and for wide temperature range requirements, it is preferable (due to equipment design limitations) and often economical to employ multi-stage reciprocating machines or centrifugal / screw compressors.

•There are two types of multi-staging systems, which are applicable to all types of compressors: compound and cascade.

Compound

•a combination of two compressors with low compression ratios can provide a high compression ratio

•A single refrigerant is used in the system, and the two compressors share the compression task equally.

•a first-stage compressor that sized to meet the cooling load, feeds into the suction of a second-stage compressor after inter-cooling of the gas. A part of the high-pressure liquid from the condenser is flashed and used for liquid sub-cooling.

•The second compressor, therefore, has to meet the load of the evaporator and the flash gas.

Cascade

- For temperatures in the range of -46°C to -101°C , cascaded systems are preferable.
- In this system, two separate systems using different refrigerants are connected so that one rejects heat to the other.
- The main advantage of this system is that a low temperature refrigerant, which has a high suction temperature and low specific volume, can be selected for the low-stage to meet very low temperature requirements.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	4. Matching Capacity to Load System • Most applications have varying loads • Consequence of part-load operation • COP increases • but lower efficiency • Match refrigeration capacity to load requires knowledge of • Compressor performance • Variations in ambient conditions • Cooling load 18 © UNEP 2006

• Consideration of part-load operation is important, because most refrigeration applications have varying loads. The load may vary due to variations in temperature and process cooling needs.

• During part-load operation, the evaporator temperature rises and the condenser temperature falls, effectively increasing the COP. But at the same time, deviation from the design operation point and the fact that mechanical losses form a greater proportion of the total power negate the effect of improved COP, resulting in lower part-load efficiency.

• Matching refrigeration capacity to the load is a difficult exercise, requiring knowledge of compressor performance, and variations in ambient conditions, and detailed knowledge of the cooling load.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	5. Capacity Control of Compressors • Cylinder unloading, vanes, valves • Reciprocating compressors: step-by-step through cylinder unloading: • Centrifugal compressors: continuous modulation through vane control • Screw compressors: sliding valves • Speed control • Reciprocating compressors: ensure lubrication system is not affected • Centrifugal compressors: >50% of capacity 19 © UNEP 2006

The capacity of compressors is controlled in a number of ways. We will discuss four of these

- **Capacity control through cylinder unloading, vanes and valves**
 - Capacity control of reciprocating compressors through cylinder unloading results in incremental (step-by-step) modulation.
 - In contrast, continuous modulation occurs in centrifugal compressors through vane control
 - and in screw compressors through sliding valves.
- **Capacity regulation through speed control** is the most efficient option.
 - Reciprocating compressors: it should be ensured that the lubrication system is not affected.
 - Centrifugal compressors: restrict speed control to about 50 % of the capacity to prevent surging. Below 50%, vane control or hot gas bypass can be used for capacity modulation.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	5. Capacity Control of Compressors • Temperature monitoring • Reciprocating compressors: return water (if varying loads), water leaving chiller (constant loads) • Centrifugal compressors: outgoing water temperature • Screw compressors: outgoing water temperature • Part load applications: screw compressors more efficient 20 © UNEP 2006

- Temperature control requires careful system design.
 - Reciprocating compressors: for widely varying loads control the compressor by monitoring the return water (or other secondary coolant) temperature.
 - If load fluctuations are not high: the temperature of the water leaving the chiller should be monitored.
 - The outgoing water temperature should be monitored for centrifugal and screw chillers.
- Capacity regulation through speed control is the most efficient option.
- The efficiency of screw compressors operating at part load is generally higher than either centrifugal compressors or reciprocating compressors, which may make them attractive in situations where part-load operation is common.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	6. Multi-Level Refrigeration Bank of compressors at central plant • Monitor cooling and chiller load: <i>1 chiller full load more efficient than 2 chillers at part-load</i> • Distribution system: individual chillers feed all branch lines; Isolation valves; Valves to isolate sections • Load individual compressors to full capacity before operating second compressor • Provide smaller capacity chiller to meet peak demands 21 © UNEP 2006

Managing part-load at bank of compressors at central plant

• Many industries use a bank of compressors at a central location to meet the load. Usually the chillers feed into a common header from which branch lines are taken to different locations in the plant. In such situations, operation at part-load requires extreme care. For efficient operation,

- the cooling load, and the load on each chiller must be monitored closely. It is more efficient to operate a single chiller at full load than to operate two chillers at part-load.
- The distribution system should be designed such that individual chillers can feed all branch lines.
- Isolation valves must be provided to ensure that chilled water (or other coolant) does not flow through chillers not in operation.
- Valves should also be provided on branch lines to isolate sections where cooling is not required. This reduces pressure drops in the system and reduces power consumption in the pumping system.
- Individual compressors should be loaded to their full capacity before operating the second compressor.
- In some cases it is economical to provide a separate smaller capacity

chiller, which can be operated on an on-off control to meet peak demands, with larger chillers meeting the base load.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	6. Multi-Level Refrigeration Packaged units (instead of central plant) • Diverse applications with wide temp range and long distance • Benefits: economical, flexible and reliable • Disadvantage: central plants use less power Flow control • Reduced flow • Operation at normal flow with shut-off periods 22 © UNEP 2006

Packaged units

- Packaged units can be used
 - For diverse applications requiring a wide range of temperatures
 - depending on the distance at which cooling loads need to be met.
- Packaged units at load centers reduce distribution losses in the system.
- Advantages: Economical, flexibility and reliability.
- Despite the advantages of packaged units, central plants generally have lower power consumption since at reduced loads power consumption can reduce significantly due to the large condenser and evaporator surfaces.

Flow control

- Flow control is also commonly used to meet varying demands.
 - Reduced flow. In such cases the savings in pumping at reduced flow should be weighed against the reduced heat transfer in coils due to reduced velocity.
 - In some cases, operation at normal flow rates, with subsequent longer periods of no-load (or shut-off) operation of the compressor, may result in larger savings.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	7. Chilled Water Storage • Chilled water storage facility with insulation • Suited only if temp variations are acceptable • Economical because • Chillers operate during low peak demand hours: reduced peak demand charges • Chillers operate at nighttime: reduced tariffs and improved COP 23 © UNEP 2006

- Depending on the nature of the load, it is economical to provide a chilled water storage facility with very good cold insulation. Also, the storage facility can be fully filled to meet the process requirements so that chillers need not be operated continuously.

- This system is usually economical if small variations in temperature are acceptable. This system has the added advantage of allowing the chillers to be operated at periods of low electricity demand to reduce peak demand charges. Low tariffs offered by some electric utilities for operation at nighttime can also be taken advantage of by using a storage facility. An added benefit is that lower ambient temperature at night lowers condenser temperature and thereby increases the COP.

- If temperature variations cannot be tolerated, it may not be economical to provide a storage facility since the secondary coolant would have to be stored at a temperature much lower than required to provide for heat gain. The additional cost of cooling to a lower temperature may offset the benefits. The solutions are case specific. For example, in some cases it may be possible to employ large heat exchangers, at a lower cost burden than low temperature chiller operation, to take advantage of the storage facility even when temperature variations are not acceptable. Ice bank systems, which store ice rather than water, are often economical.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment/ Refrigeration & AC 	8. System Design Features • FRP impellers, film fills, PVC drift eliminators • Softened water for condensers • Economic insulation thickness • Roof coatings and false ceilings • Energy efficient heat recovery devices • Variable air volume systems • Sun film application for heat reflection • Optimizing lighting loads 24 © UNEP 2006

• In overall plant design, adoption of good practices improves the energy efficiency significantly. Some areas for consideration are:

- Design of cooling towers with FRP impellers and film fills, PVC drift eliminators, etc.
- Use of softened water for condensers in place of raw water.
- Use of economic insulation thickness on cold lines, heat exchangers, considering cost of heat gains and adopting practices like infrared thermography for monitoring - applicable especially in large chemical / fertilizer / process industry.
- Adoption of roof coatings / cooling systems, false ceilings / as applicable, to minimize refrigeration load.
- Adoption of energy efficient heat recovery devices like air to air heat exchangers to pre-cool the fresh air by indirect heat exchange; control of relative humidity through indirect heat exchange rather than use of duct heaters after chilling.
- Adopting of variable air volume systems
- Adopting of sun film application for heat reflection
- Optimizing lighting loads in the air conditioned areas
- Optimizing number of air changes in the air conditioned areas are few

other examples.

 UNEP	Training Session on Energy Equipment
Electrical Equipment/ Refrigeration & AC Gerlap	Refrigeration & Air Conditioning Systems THANK YOU FOR YOUR ATTENTION 25 © UNEP 2006

Thank you for your attention!